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DATSECURE LIMITED NETWORK SIMULATION AND SECURITY IMPLEMENTATION REPORT

DatSecure Limited is a Luxembourg-based data processing company that was established in 2023. The business specializes in handling and protecting sensitive data, especially in the healthcare sector, where it helps hospitals and health insurance companies share secure data. DatSecure Limited has to make sure that its network infrastructure is safe, scalable, and effective as part of its recent North American development. Cisco Packet Tracer was used to create a virtual office network in order to help achieve this goal. This enables the business to test settings, find security flaws, and put security measures in place before implementing them in a live system.

As a Junior Network Engineer and Cybersecurity Analyst for this project, I was in charge of network design, connectivity verification, and the implementation of Access Control Lists (ACLs) to protect departmental communication.

Business Problem and Objectives

The Problem

For its new headquarters, DatSecure Limited needed a network design that was both safe and effective. The main challenge was to:

- Facilitate departmental communication
- Limit unwanted access to private information
- To prevent expensive mistakes in the real world, simulate the network environment.

Project Objectives

This project's primary goals were to:

- Create a network with many departments using switches and a router.
- Set up IP addresses for every department.
- Use simple networking tools, such as ping, to confirm connectivity.
- Use ACLs to manage and limit network traffic.
- Verify and test service availability and access control.

Network Design and Configuration

Three organizational departments were represented by the network's structure (Admin, Sales and HR) with two devices (PCs) each. Admin was configured to have full access level, Sales has limited access and HR has restricted access (internet only).

The Network Components was laid out such that;

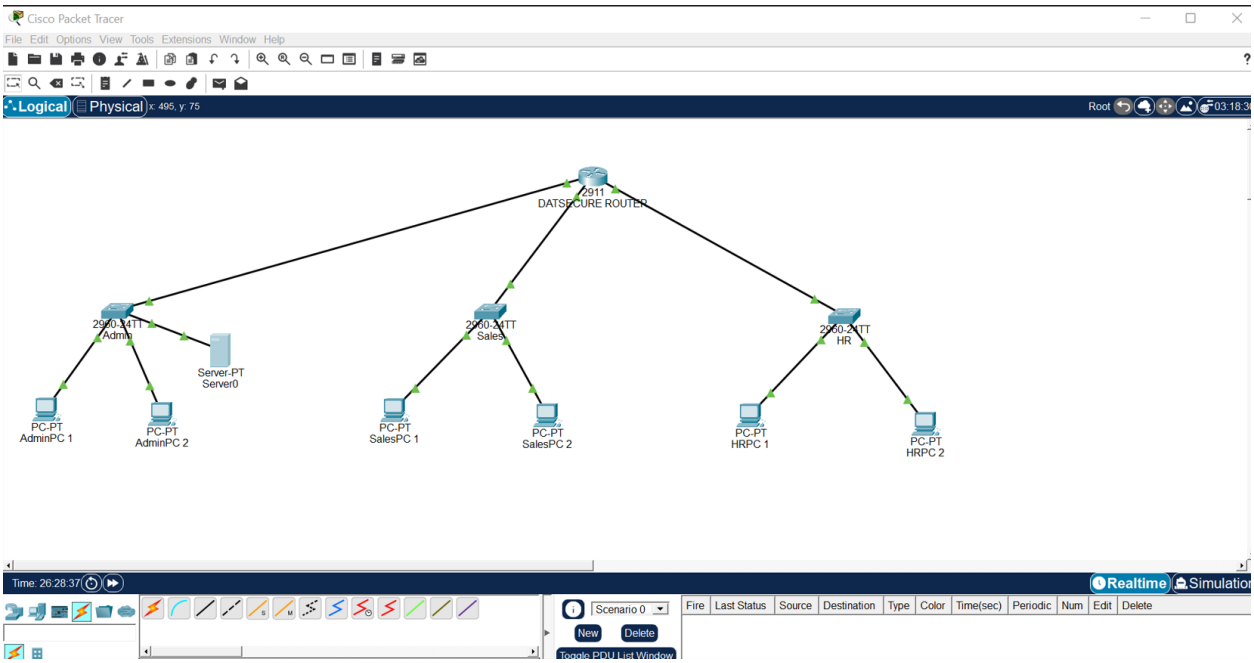
All departmental networks are connected by a router (DatSecure router).

Each department has one switch.

PCs: Used to imitate consumer devices

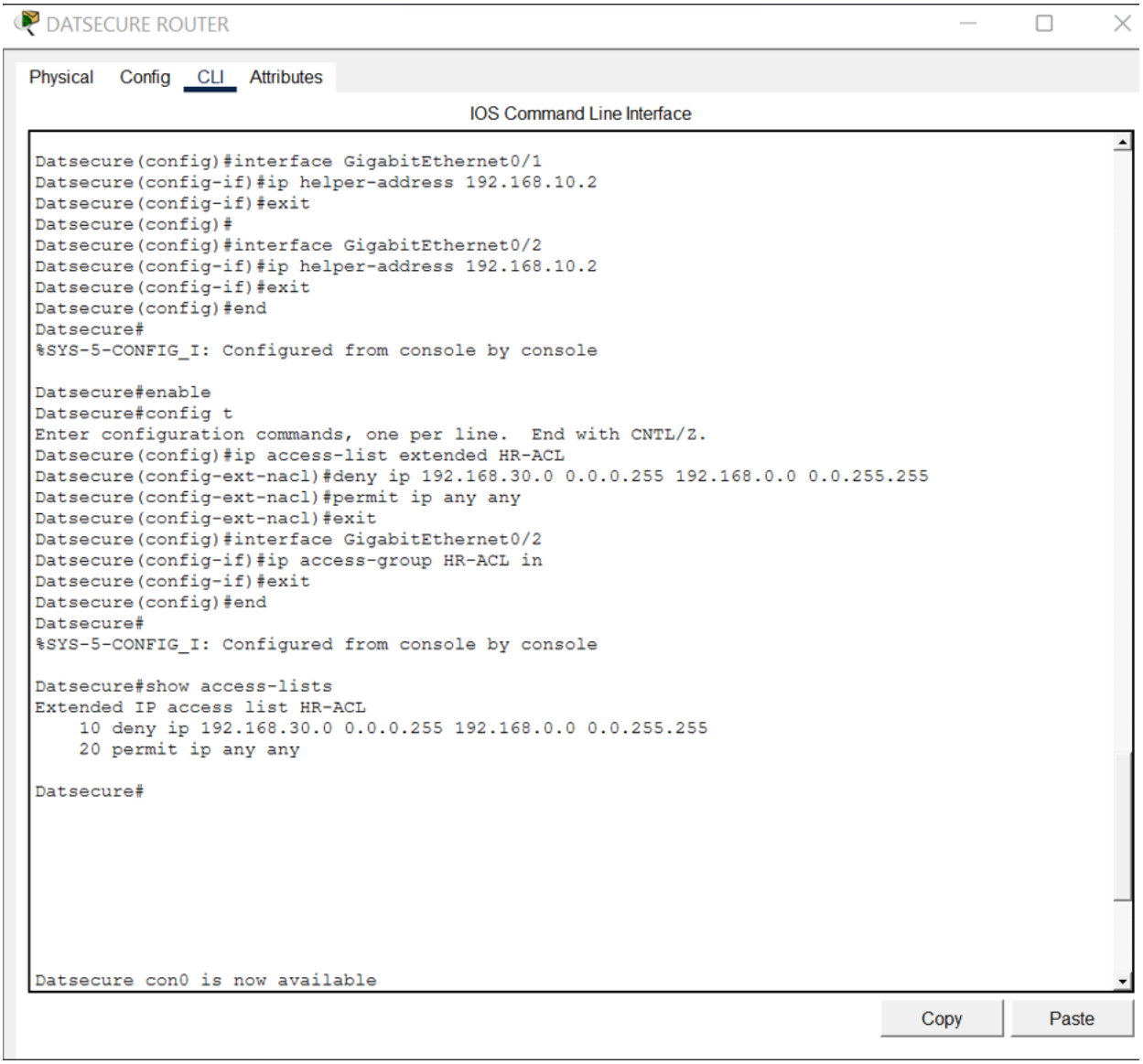
Server: Located on the administrative network

Below is a visual representation and layout of the Office Network:



Better departmental control and security are ensured by this division. This segmentation ensures better control and security across departments. As part of the task, a server was also added to the Admin network. ACLs were set up on the router to control server access in order to improve network security. The security requirements were that Admins and Sales were allowed access into the Server, while HR was completely denied access. It was configured to make sure that HR was prohibited from using HTTP to access the server, while only the administrative and sales units could.

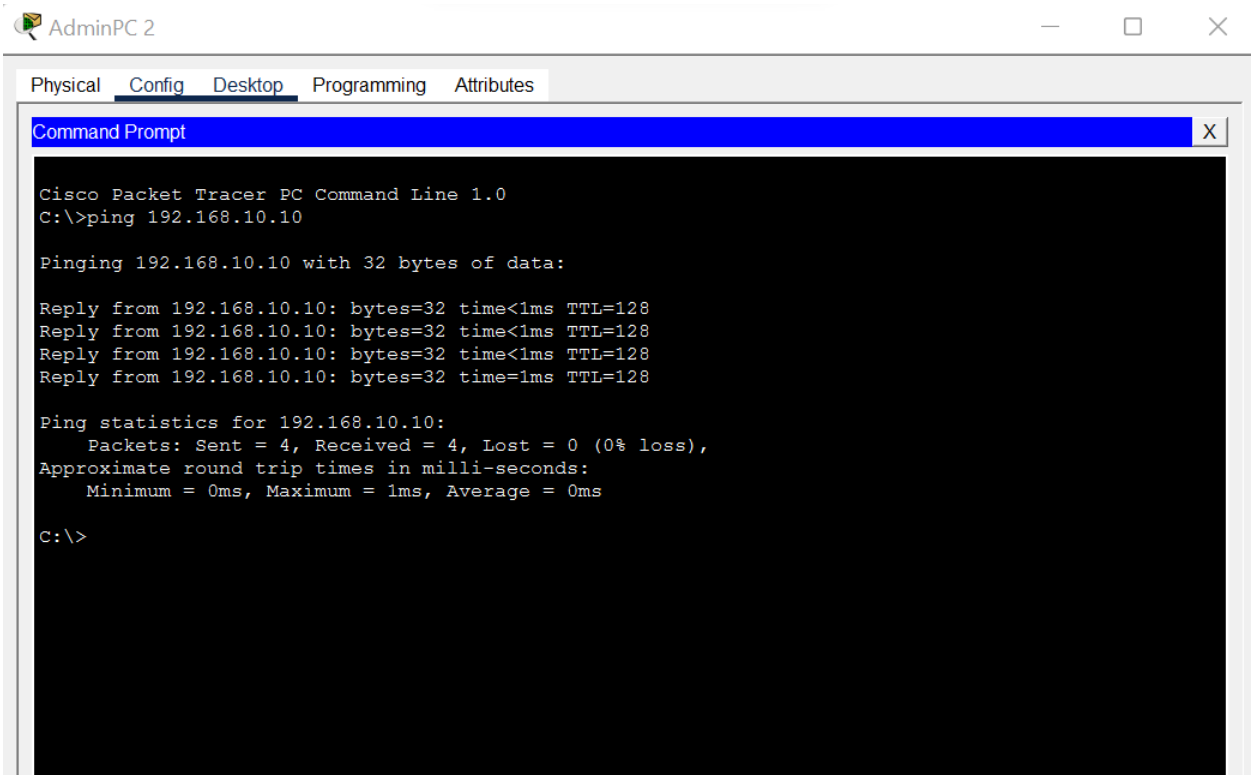
Below is a picture of the ACL Configuration commands:

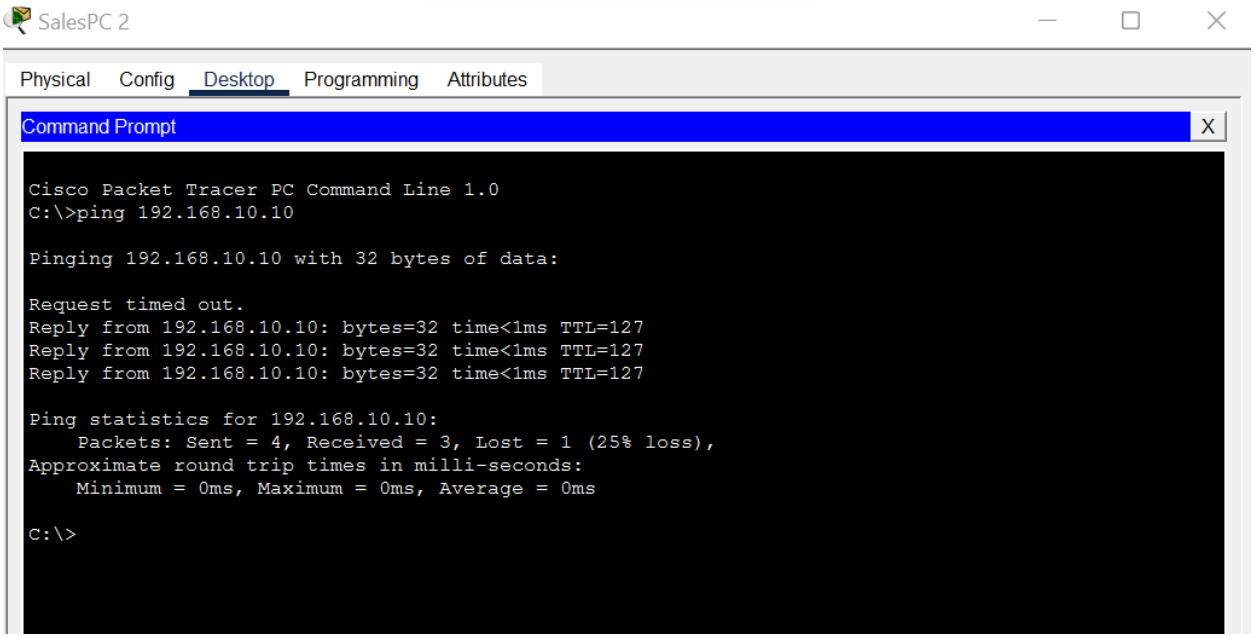


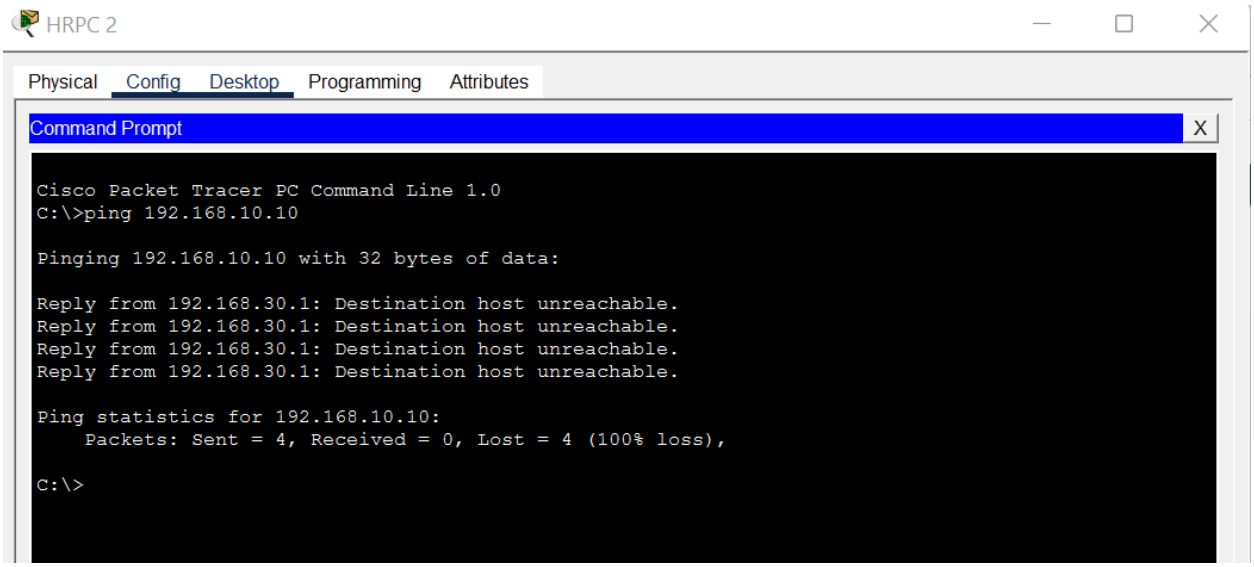
Testing and Verification

Tests were carried out to verify proper functionality after the network and ACLs were configured. Within their respective networks, devices were able to communicate with each other. Inter-network connectivity was verified using ping tests. Ping and server access were possible for the admin and sales PCs. HR PCs were unable to successfully connect to the server.

Below are images of the PC connection checks to the server using ping:







This assignment illustrated how ACLs may be used to improve network security in a controlled setting. By applying access controls and dividing the network, it ensured resources that were sensitive were safeguarded. Unauthorized access was also stopped, and network traffic was efficiently controlled. For businesses like DatSecure Limited that deal with sensitive healthcare data, where security and privacy are crucial, this strategy is even more important.

In summary up, the project accomplished all of its goals. Cisco Packet Tracer was used to design and implement a safe and useful network. The system was set up such that only approved departments could access it, thereby limiting HR's access.

In order to prepare for real-world deployment, our simulation offered insightful information on network configuration, design, and security implementation. It also emphasizes how crucial ACLs are to preserving safe and effective network operations.